23AID498 – Project Phase 1

Project Diary

Student Name:	Harita Aneesh
Roll No:	AM.AI.U4AID23042
Team No:	04
Guide Name:	Dr. Divya Udayan J
Project Title:	Optimizing Autonomous Traffic Navigation via Reinforcement Learning and Basal Ganglia Gating in ROS2

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 14/07/2026

Work planned for the coming week	• Identify and instantiate ego vehicle as the central autonomous agent within the SUMO network. • Implement a bidirectional communication link between simulation environment and controller • Setup motor control feedback loop & spatial tracking mechanism
Work completed during the previous week	• Integrated SUMO environment • Setup the periodic traffic light controller. • Setup the SUMO vehicle control • Setup the bidirectional road lane.
Pending work	• Need to setup PPO-policy action • Need to integrate a monitoring system to detect the traffic light phases

Remarks by the Guide: Progressing.

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 16/07/2026.

Work planned for the coming week	• Need to enable hippocampal module to map vehicle's position. • Need to integrate a safety critical control.
Work completed during the previous week	Literature review • Collected 10 journals and compared with our project. • Followed by novelty identification and phase I phase II division. set motor control
Pending work	

Remarks by the Guide:

Progressing.

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 22/07/2026

Work planned for the coming week	• Planning to make the model dynamic • Increase the complexity of the model.
Work completed during the previous week	• Enable hippocampal module to map vehicle's position • Integrated a safety critical control • Drafted Literature - PPT.
Pending work	

Remarks by the Guide: ppt presented

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: 22/7/26

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 30/07/2026

Work planned for the coming week	◦ Increase the time limit of model execution. ◦ Improve the Action Selection process by adding SAC.
Work completed during the previous week	◦ Increased the complexity of model by increasing traffic and adjusted the signal simulation ◦ Drafted System Design
Pending work	

Remarks by the Guide: Progressing.

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: Dr. Divya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 07/08/2026

Work planned for the coming week	• Fine tune traffic density thresholds and deadlock wait timers based on extended continuous runs • Begin mapping out a larger, multi-intersection urban network to test generalization.
Work completed during the previous week	• Implemented time-varying dynamic background traffic to simulate rush hour flows • Integrated aggressive, heavy, erratic vehicles into the SUMO network.
Pending work	• Need to develop automated deadlock resolution to clear trapped vehicles and force faster signal cycles.

Remarks by the Guide:

Progressing

Weekly Performance Rating (0–5): 4.5/5

Guide's Signature with Date:

Dr. Diya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 04/09/2026

Work planned for the coming week	Transition from standard to hybrid SAC architecture, optimize policy resilience for high traffic and chaotic scenarios, and lay the groundwork for ROS2 network integration.
Work completed during the previous week	Standard SAC training, policy evaluation for the hybrid transition and defined ROS2 /cmd_vel requirements.
Pending work	finalize hybrid SAC transition, continue chaotic traffic training, and begin ROS2 /cmd_vel testing.

Remarks by the Guide: Progressing -

4.5

Weekly Performance Rating (0–5): ___ / 5

Dr. Diya Udayent

Guide's Signature with Date:

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 09/09/2026

Work planned for the coming week	Connect hybrid SAC to ROS2, test /cmd_vel transmission, run RQ3 traffic-density evaluations and prepare for hardware deployment
Work completed during the previous week	Upgraded to hybrid SAC, trained for chaotic traffic, and prepared ROS2 integration components.
Pending work	Complete ROS2 integration, finish RQ3 evaluations and conduct physical hardware testing.

Remarks by the Guide: Progressing.

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: 09/9/26

Dr. Dimple Udyawar

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 16/09/2025.

Work planned for the coming week	• Initial code cleanup & structural organisation for Sumo environment • Verify & expand the literature survey bibliography to include full author lists and publication years
Work completed during the previous week	• Developed Sumo Traffic environment using Traci, successfully disabled lane car following model. • Designed & implemented multi objective reward function • Integrated exponential safety potential field.
Pending work	• Drafting the initial layout for the empirical benchmark matrices matrices & gathering literature survey data.

Remarks by the Guide: Code review discussed.

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: Dr. Diya Udayan

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 24/09/2026.

Work planned for the coming week	conduct final validation runs for the proposed TD3+ BiO shield architecture to strictly verify the collision rate under high density procedural traffic
Work completed during the previous week	• Formatted the empirical benchmark tables in LaTeX, resolving float spacing issues by merging the unshielded / shielded matrices
Pending work	• final code cleanup & structural organization for the SUMO environment Python scripts. • Resolving any residual margin or overfull \hbox warnings in the compiled LaTeX manuscript

Remarks by the Guide: Progressing.

Weekly Performance Rating (0–5): 4 5 / 5

Guide's Signature with Date: Dr. Divya Udayan J

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 08/10/26

Work planned for the coming week	• Prepare the SumoTraffic tw and action space mapping to support the upcoming integration of DDPG algorithm.
Work completed during the previous week	• Code review evaluation completed. • Conducted final validation runs for the proposed TD3+ Bio-shield architecture, strictly verifying 0.0% collision rate
Pending work	• Drafting the slide deck and visual assets for the final project presentation and defense.

Remarks by the Guide:

Progressing.

4.5

Weekly Performance Rating (0–5): ____ / 5

Guide's Signature with Date:

Dr. Divya

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 13|10|'26.

Work planned for the coming week	• Run ablation episodes assessing the impact of sensor observation noise on policy stability. • Optimize Traci batch command execution to accelerate simulation steps during multi-epoch policy evaluations.
Work completed during the previous week	• Extended the SumoTrafficEnv observation and step pipeline to expose deterministic grad. feedback. • Calibrated throttle-to-acceleration transfer functions to prevent extreme gradient saturation
Pending work	• Resolving step-synchronization latency between Traci physics updates and actor gradient backward passes.

Remarks by the Guide:

Progressing

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date:

Dr. Divya Udayan J

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 23/10/2026.

Work planned for the coming week	• Need to write the report. • Comparison of all the models made till now.
Work completed during the previous week	• Prepared the sumo Traffic Env and action space mapping to support the upcoming integration of ddpg algorithm. • Made the conference paper.
Pending work	

Remarks by the Guide: Progress.

Weekly Performance Rating (0–5): 4.5 / 5

Guide's Signature with Date: Dr. Divya Udayan